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Report Concerning Space Data System Standards

OVERVIEW OF SPACE COMMUNICATIONS PROTOCOLS

INFORMATIONAL REPORT

CCSDS 130.0-G-3

GREEN BOOK
July 2014



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CCSDS Secretariat
National Aeronautics and Space Administration
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E-mail: secretariat@mailman.ccsds.org

FOREWORD

This document is a CCSDS Report that contains an overview of the space communications protocols recommended by CCSDS. A space link is a communications link between a spacecraft and its associated ground system or between two spacecraft. A space communications protocol is a communications protocol designed to be used over a space link, or in a network that contains one or multiple space links.

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1 INTRODUCTION

1.1 PURPOSE AND SCOPE

The purpose of this Report is to provide an architectural overview of the space communications protocols recommended by CCSDS and to show how these protocols are used in space mission data systems.

A space link is a communications link between a spacecraft and its associated ground system or between two spacecraft. A space communications protocol is a communications protocol designed to be used over a space link, or in a network that contains one or multiple space links.

This Report presents only a top-level overview of the space communications protocols and does not contain the specification or rationale of each protocol. The specification of a space communications protocol developed by CCSDS is contained in a CCSDS Blue Book, and its rationale is described in a CCSDS Green Book that accompanies the Blue Book.

1.2 DOCUMENT STRUCTURE

This document is divided into four numbered sections and an annex:

- a) section 1 presents the purpose and scope of this Report and lists the definitions and references used throughout the Report;
- b) section 2 provides a brief introduction to the space communications protocols;
- c) section 3 presents major features of the space communications protocols;
- d) section 4 shows some examples of how space communications protocols are used in space data systems;
- e) annex A lists acronyms and abbreviations used within this document.

1.3 DEFINITIONS

1.3.1 DEFINITIONS FROM OSI BASIC REFERENCE MODEL

Most of the CCSDS space communications protocols are defined using the style established by the Open Systems Interconnection (OSI) Basic Reference Model (reference [2]). This model provides a common framework for the development of standards in the field of systems interconnection. It defines concepts and terms associated with a layered architecture and introduces seven specific layers. The concepts and terms defined in this model are extensively used in the Blue Books that define CCSDS space communications protocols. If the reader is not familiar with this model, an excellent introduction can be found in a textbook on computer networks such as reference [3].

The following terms used in this Report are defined in reference [2]:

- a) Application Layer;
- b) Data Link Layer;
- c) layer;
- d) Network Layer;
- e) Physical Layer;
- f) protocol data unit;
- g) service;
- h) Transport Layer.

1.3.2 TERMS DEFINED IN THIS REPORT

For the purposes of this Report, the following definitions also apply.

forwarding: The act of transferring data from its source towards its destination, which may be in space or on the ground.

octet: An 8-bit word.

Physical Channel: A stream of bits transferred over a space link (see below) in a single direction.

routing: The process of selecting paths from origins to destinations in a network.

space link: A communications link between a spacecraft and its associated ground system or between two spacecraft. A space link consists of one or more Physical Channels in one or both directions.

space communications protocol: A communications protocol designed to be used over a space link (see above), or in a network that contains one or multiple space links.

1.4 REFERENCES

The following documents are referenced in this Report. At the time of publication, the editions indicated were valid. All documents are subject to revision, and users of this Report are encouraged to investigate the possibility of applying the most recent editions of the documents indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS documents.

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2 INTRODUCTION TO SPACE COMMUNICATIONS PROTOCOLS

2.1 HISTORY OF SPACE COMMUNICATIONS PROTOCOLS

Traditionally, telemetry transmitted from the spacecraft was formatted with a Time Division Multiplexing (TDM) scheme, where data items were multiplexed into a continuous stream of fixed-length frames based on a predefined multiplexing rule. To design and implement a data system for spacecraft, each project was forced to develop a custom system used by that project alone, with the exception of the ground tracking network, because of the lack of established standards in this field.

The advent of microprocessor-based spacecraft instruments and subsystems, however, enabled telemetry systems to become more flexible and have greater throughput so that data processed by onboard software could be transmitted efficiently.

In the early 1980s, CCSDS developed an international standard for a Packet Telemetry protocol capable of sending processed telemetry efficiently using a variable-length data unit called the Source Packet. Source Packets generated by various instruments and subsystems on a spacecraft are transmitted to the ground in a stream of continuous, fixed-length Transfer Frames. This standard has been used by many space projects enabling them to share onboard and ground data processing equipment.

Based on a similar concept, another international standard on Telecommand was developed by CCSDS, shortly after Packet Telemetry, for sending commands to a spacecraft with a data unit known as the TC Packet. TC Packets destined for various instruments and subsystems on a spacecraft are transmitted from the ground in a stream of sporadic, variable-length Transfer Frames.

In the late 1980s, CCSDS extended the above standards to meet the requirements of the Advanced Orbiting Systems, such as the International Space Station, and came up with a third standard known as AOS. The AOS standard added to the Packet Telemetry standard services for transmitting various types of online data (such as audio and video data). And it may be used on both space-to-ground and ground-to-space links. The AOS uses the same packet structure as the Packet Telemetry standard but the frame format is slightly different.

These three standards (Packet Telemetry, Telecommand, and AOS) were later restructured by CCSDS in order to define the protocols in a more structured and unified way, and the following standards replaced the original standards:

- a) Space Packet Protocol (reference [4]);
- b) TM, TC and AOS Space Data Link Protocols (references [5], [6], and [7], respectively);
- c) TM and TC Synchronization and Channel Coding (references [8] and [9], respectively).

As an international standard for the Radio Frequency (RF) signal between a spacecraft and a ground station, CCSDS developed a standard called Radio Frequency and Modulation Systems (reference [10]). This standard specifies the characteristics of the RF signal used to carry Packets and Frames.

In the 1990s, CCSDS developed another set of protocols collectively known as Space Communications Protocol Specifications (SCPS), which include SCPS Network Protocol (SCPS-NP) (reference [11]), SCPS Security Protocol (SCPS-SP) (reference [12]), SCPS Transport Protocol (SCPS-TP) (reference [13]), and SCPS File Protocol (SCPS-FP) (reference [14]). The SCPS protocols are generally based on Internet protocols. But modifications and extensions to the Internet protocols are incorporated in the design of the SCPS protocols to meet the specific needs of space missions. CCSDS has retired all of the SCPS protocols with the exception of SCPS-TP.

In response to the needs of space missions to transfer files to and from an onboard mass memory, CCSDS has developed a protocol called the CCSDS File Delivery Protocol (CFDP) (reference [15]). This protocol provides the capability to transfer files reliably and efficiently over an unreliable protocol (for example, the Space Packet Protocol).

In the area of data compression, CCSDS has developed a Lossless Data Compression standard (reference [16]) and an Image Data Compression standard (reference [17]) either to increase the science return or to reduce the requirement for onboard memory, station contact time, and data archival volume. The former standard guarantees full reconstruction of the original data without incurring any distortion in the process, while with the latter standard, quantization or other approximations used in the compression process may result in the inability to reproduce the original data set without some distortion.

Recently CCSDS has developed a protocol called Proximity-1 Space Link Protocol (references [18], [19], [20], and [32]) to be used over proximity space links. Proximity space links are defined to be short range, bi-directional, fixed or mobile radio links, generally used to communicate among fixed probes, landers, rovers, orbiting constellations, and orbiting relays. This protocol defines a data link protocol (reference [18]), coding and synchronization methods (reference [19]), and RF and modulation characteristics (reference [20]).

Security is of great concern to many space missions. CCSDS has published several documents, including *The Application of CCSDS Protocols to Secure Systems* (reference [37]), *Security Architecture for Space Data Systems* (reference [53]), and *CCSDS Cryptographic Algorithms* (reference [54]) to provide guidance to missions that wish to use the CCSDS space communications protocols for spacecraft control and data handling but also require a level of security or data protection.

2.2 PROTOCOL LAYERS

2.2.1 SUMMARY

A communications protocol is usually associated with one of the seven layers defined in the OSI Basic Reference Model (reference [2]). Although some space communications protocols do not fit well with the OSI seven-layer model, this Report uses this model for categorizing the space communications protocols.

The space communications protocols are defined for the following five layers of the ISO model:

- a) Physical Layer;
- b) Data Link Layer;
- c) Network Layer;
- d) Transport Layer;
- e) Application Layer.

As in most terrestrial networks, protocols of the Session and Presentation Layers of the OSI model are rarely used over space links.

Figure 2-1 shows the space communications protocols categorized into the five layers listed above. Figure 2-2 shows some possible combinations of these protocols.

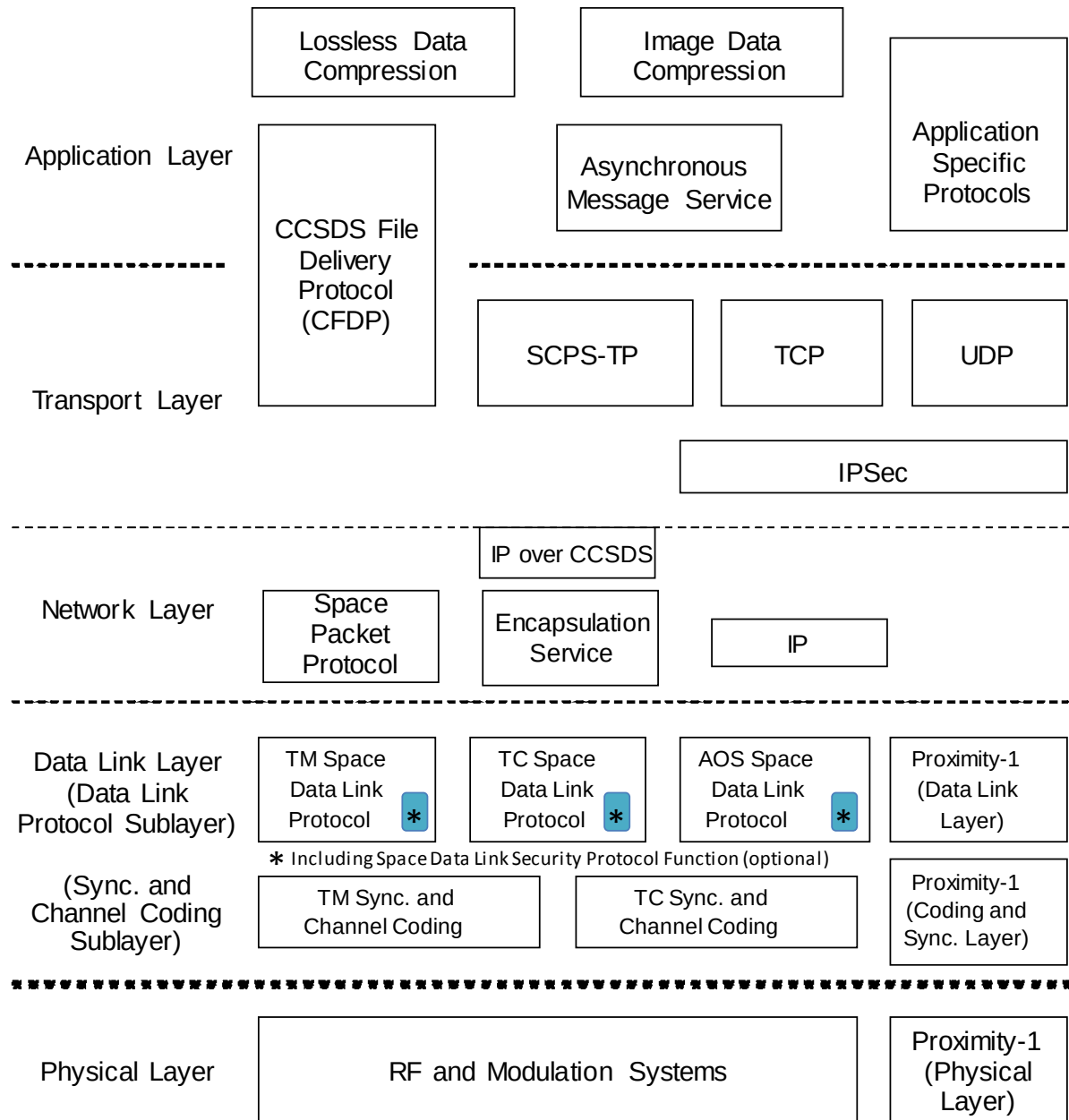
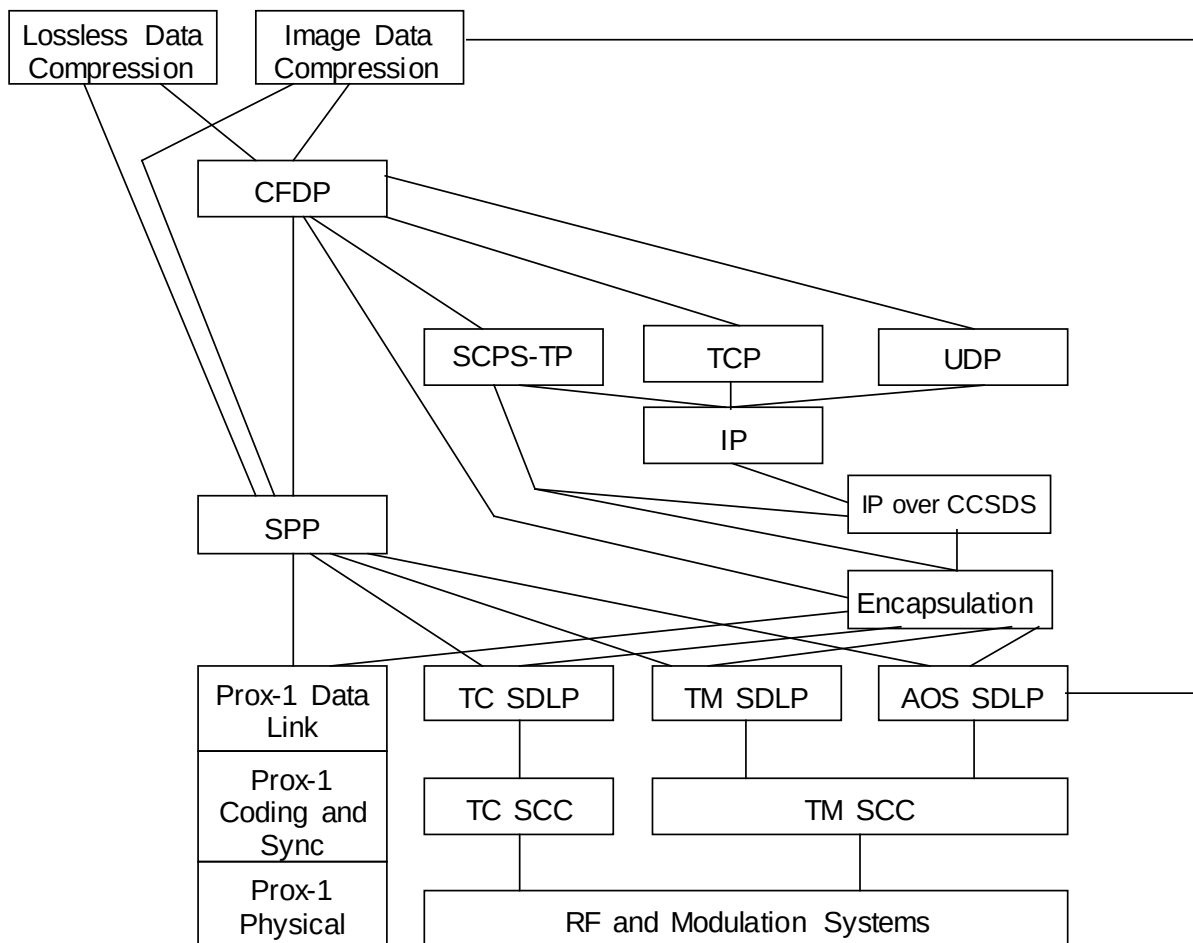


Figure 2-1: Space Communications Protocols Reference Model



SCPS-SP and IPsec can be used between the Transport and Network layers in any combination of protocols.

SPP = Space Packet Protocol

SDLP = Space Data Link Protocol & Space Data Link Security (opt.)

SCC = Synchronization and Channel Coding

Figure 2-2: Some Possible Combinations of Space Communications Protocols

In figure 2-1, there are two protocols that do not correspond to a single layer. CCSDS File Delivery Protocol (CFDP) has the functionality of the Transport and Application Layers. Proximity-1 Space Link Protocol has the functionality of the Data Link and Physical Layers.

CCSDS does not formally define Application Program Interfaces (APIs) for the space communications protocols, but most CCSDS standards provide abstract service definitions in the form of primitives following the conventions established by ISO (see reference [21]). A primitive is an abstract representation of the services provided by the protocol layer, but it does not depend on any implementation technology. This abstract specification may be used as a reference for developing an API.

In the following subsections, the protocols shown in figure 2-1 are briefly introduced. Major features of these protocols will be explained in section 3.

2.2.2 PHYSICAL LAYER

CCSDS has an omnibus standard for the Physical Layer called the Radio Frequency and Modulation Systems (reference [10]) to be used for space links between spacecraft and ground stations. The Proximity-1 Space Link Protocol also contains recommendations for the Physical Layer of proximity space links (reference [20]).

2.2.3 DATA LINK LAYER

CCSDS defines two Sublayers in the Data Link Layer of the OSI Model: Data Link Protocol Sublayer and Synchronization and Channel Coding Sublayer. The Data Link Protocol Sublayer specifies methods of transferring data units provided by the higher layer over a point-to-point space link using data units known as Transfer Frames. The Synchronization and Channel Coding Sublayer specifies methods of synchronization and channel coding for transferring Transfer Frames over a space link.

CCSDS has developed four protocols for the Data Link Protocol Sublayer of the Data Link Layer:

- a) TM Space Data Link Protocol (reference [5]);
- b) TC Space Data Link Protocol (reference [6]);
- c) AOS Space Data Link Protocol (reference [7]);
- d) Proximity-1 Space Link Protocol—Data Link Layer (reference [18]).

The above protocols provide the capability to send data over a single space link. TM, TC, and AOS have provision for inserting secured user data into a frame using the Space Data Link Security (SDLS) Protocol (reference [43]). However, there have been no security requirements to date established for Proximity-1. The SDLS protocol can provide security services, such as authentication and confidentiality, for TM Transfer Frames, AOS Transfer Frames, and/or TC Transfer Frames. Note that the use of the SDLS function within these protocols is optional.

CCSDS has developed three standards for the Synchronization and Channel Coding Sublayer of the Data Link Layer:

- a) TM Synchronization and Channel Coding (reference [8]);
- b) TC Synchronization and Channel Coding (reference [9]);
- c) Proximity-1 Space Link Protocol—Coding and Synchronization Layer (reference [19]).

TM Synchronization and Channel Coding is used with the TM or AOS Space Data Link Protocol, TC Synchronization and Channel Coding is used with the TC Space Data Link

Protocol, and the Proximity-1 Space Link Protocol—Coding and Synchronization Layer is used with the Proximity-1 Space Link Protocol—Data Link Layer.

The TM, TC and AOS Space Data Link Protocols and the Proximity-1 Space Link Protocol (Data Link Layer) are called the Space Data Link Protocols in this document.

2.2.4 NETWORK LAYER

Space communications protocols of the Network Layer provide the function of routing or forwarding higher-layer data through the entire data system that includes both onboard and ground subnetworks.

CCSDS has two standards for interfacing at the Network Layer:

- a) Space Packet Protocol (reference [4]);
- b) Encapsulation Service (reference [29]).

For the Space Packet Protocol, Protocol Data Units (PDUs) are generated and consumed by application processes that are on a spacecraft or on the ground.

CCSDS Encapsulation Service allows encapsulation of PDUs of CCSDS recognized protocols, as defined in a SANA registry (reference [47]) into one of two types of CCSDS packets i.e., either Space Packets or Encapsulation Packets. These packets can then be transferred by CCSDS Space Data Link Protocols over a space link.

CCSDS-recognized Internet datagrams (listed in reference [44]) can also be transferred by CCSDS Space Data Link Protocols over a space link, multiplexed or not-multiplexed, using IP over CCSDS (reference [45]).

2.2.5 TRANSPORT LAYER

Space communications protocols of the Transport Layer provide users with end-to-end transport services.

CCSDS has developed the SCPS Transport Protocol (SCPS-TP) (reference [13]) for the Transport Layer.

PDUs of a Transport Layer protocol are usually transferred with a protocol of the Network Layer over a space link, but they can be transferred directly by a Space Data Link.

Transport protocols used in the Internet (such as TCP, reference [24], and UDP, reference [25]) can also be used on top of IP datagrams over CCSDS space links, reference [45]. IPsec (reference [27]) may be used with a Transport protocol of the Internet suite to provide end-to-end data protection capability.

2.2.6 APPLICATION LAYER

Space communications protocols of the Application Layer provide users with end-to-end application services such as file transfer and data compression.

CCSDS has developed five protocols for the Application Layer:

- a) Asynchronous Messaging Service (AMS) (reference [46]);
- b) CCSDS File Delivery Protocol (CFDP) (reference [15]);
- c) Lossless Data Compression (reference [16]);
- d) Image Data Compression (reference [17]);
- e) Lossless Multispectral & Hyperspectral Image Compression (reference [48]).

AMS is an application layer protocol for end-to-end mission data system message transfer.

CFDP provides the functionality of the Application Layer (i.e., functions for file management). The CFDP Store-and-Forward Overlay procedures provide application-specific transfer of data across multiple link-layer hops.

Each project (or Agency) may also elect to use application-specific protocols not recommended by CCSDS to fulfill their mission requirements in the Application Layer over CCSDS space communications protocols.

PDUs of an Application Layer protocol are usually transferred with a protocol of the Transport Layer over a space link, but they can be transferred directly with a protocol of the Network Layer.

Applications protocols used in the Internet (such as FTP, reference [26]) can also be used on top of SCPS-TP, TCP and UDP over space links.

3 MAJOR FEATURES OF SPACE COMMUNICATIONS PROTOCOLS

3.1 PHYSICAL LAYER

The CCSDS Recommendation Standard for Radio Frequency and Modulation Systems (reference [10]) recommends the characteristics of the RF and modulation systems used for communications over space links between spacecraft and ground stations.

The Proximity-1 Space Link Protocol—Physical Layer (reference [20]) also contains recommendations for the Physical Layer of proximity space links.

3.2 DATA LINK LAYER

3.2.1 GENERAL FEATURES OF DATA LINK PROTOCOLS

CCSDS has developed four protocols for the Data Link Protocol Sublayer of the Data Link Layer:

- a) TM Space Data Link Protocol (reference [5]);
- b) TC Space Data Link Protocol (reference [6]);
- c) AOS Space Data Link Protocol (reference [7]);
- d) Data Link Protocol Sublayer portion of Proximity-1 Space Link Protocol (reference [18]).

These protocols (collectively known as Space Data Link Protocols) provide the capability to transfer various types of data on space links, but their principal function is to transfer variable-length data units known as packets.

Each packet format transferred by the Space Data Link Protocols must have a Packet Version Number (PVN) recognized by CCSDS. These numbers are contained in SANA (reference [28]). Packets with authorized Packet Version Numbers can be transferred by the Space Data Link Protocols directly, but CCSDS has another mechanism to transfer PDUs of CCSDS and non-CCSDS protocols with a service called the Encapsulation Service, defined in reference [29]. With this service, packets are transferred by the Space Data Link Protocols encapsulated in either Space Packets defined in reference [4] or Encapsulation Packets defined in reference [29].

The TM Space Data Link Protocol is usually used for (but not limited to) sending telemetry from a spacecraft to a ground station (i.e., on a return link). The TC Space Data Link Protocol is usually used for (but not limited to) sending commands from a ground station to a spacecraft (i.e., on a forward link). The AOS Space Data Link Protocol may be used on a return link alone, or on both forward and return links if there is a need for two-way higher-speed communications (e.g., audio and video) between a spacecraft and the ground. The

Proximity-1 Space Link Protocol is to be used over proximity space links, where proximity space links are defined to be short range, bi-directional, fixed or mobile radio links, generally used to communicate among fixed probes, landers, rovers, orbiting constellations, and orbiting relays.

The protocol data units used by the Space Data Link Protocols are called Transfer Frames. The TM and AOS Space Data Link Protocols use fixed-length Transfer Frames to facilitate robust synchronization procedures over a noisy link, while the TC Space Data Link Protocol and the Proximity-1 Space Link Protocol use variable-length Transfer Frames to facilitate reception of short messages with a short delay.

A key feature of all the Space Data Link Protocols is the concept of ‘Virtual Channels’. The Virtual Channel facility allows one Physical Channel (a stream of bits transferred over a space link in a single direction) to be shared among multiple higher-layer data streams, each of which may have different service requirements. A single Physical Channel may therefore be divided into several separate logical data channels, each known as a Virtual Channel (VC). Each Transfer Frame transferred over a Physical Channel belongs to one of the Virtual Channels of the Physical Channel.

The TC Space Data Link Protocol has a function for retransmitting lost or corrupted data to ensure delivery of data in sequence without gaps or duplication over a space link. This function is provided by a retransmission control mechanism called the Communications Operation Procedure-1 (COP-1), which is defined in a separate document (reference [30]). (This function does not necessarily guarantee end-to-end complete delivery.) The Proximity-1 Space Link Protocol also has a similar function called COP-P, which is defined in the Data Link Layer Recommended Standard (reference [18]). Neither the TM Space Data Link Protocol nor the AOS Space Data Link Protocol has such a function, so retransmission must be done by a higher-layer protocol if complete delivery of data is required.

The TM and AOS Space Data Link Protocols should be used together with the TM Synchronization and Channel Coding Recommended Standard (reference [8]). The TC Space Data Link Protocols should be used together with the TC Synchronization and Channel Coding Recommended Standard (reference [9]). The TM Channel Coding Profiles Magenta book (reference [51]) provides guidance to users on the use of the various coding schemes for telemetry links in different mission profiles. The TM, TC, and AOS Space Data Link Protocols should be used on top of the Recommended Standard for Radio Frequency and Modulation Systems (reference [10]).

The Proximity-1 Space Link Protocol—Data Link Layer (reference [18]) should be used together with the Proximity-1 Space Link Protocol—Data Coding and Synchronization Layer (reference [19]) and on top of the Proximity-1 Space Link Protocol—Physical Layer (reference [20]).

A summary of concept and rationale of the TM, TC and AOS Space Data Link Protocols is contained in reference [31]. A summary of rationale, architecture and scenarios of the Proximity-1 Space Link Protocol is contained in reference [32].

3.2.2 IDENTIFIERS USED BY DATA LINK PROTOCOLS

The Space Data Link Protocols provide link identifiers to identify data streams. The identifier names as well as their values reside in the Space Assigned Numbers Authority (SANA). SANA is the registrar for all protocol registries created under CCSDS. SANA replaces the retired Space Link Identifiers Blue Book. The CCSDS Global Spacecraft Identification Field: Code Assignment Control Procedures Blue Book (reference [52]) contains the procedures governing requesting, assigning, and relinquishing CCSDS SCID field codes.

The TM, TC and AOS Space Data Link Protocols have the following three identifiers: the Transfer Frame Version Number (TFVN), the Spacecraft Identifier (SCID), and the Virtual Channel Identifier (VCID).

The Transfer Frame Version Number (TFVN) is used to distinguish among different Transfer Frames. However, different Transfer Frames must not be multiplexed on a Physical Channel.

The concatenation of a TFVN and a SCID is known as a Master Channel Identifier (MCID), which is used for identifying a spacecraft associated with a space link.

All Transfer Frames with the same MCID on a Physical Channel constitute a Master Channel (MC). A Master Channel consists of one or more Virtual Channels, each of which is identified with a VCID. In most cases, a Physical Channel carries only Transfer Frames of a single MCID, and the Master Channel will be identical with the Physical Channel. However, a Physical Channel may carry Transfer Frames with multiple MCIDs (with the same TFVN). In such a case, the Physical Channel consists of multiple Master Channels. A Physical Channel is identified with a Physical Channel Name, which is set by management and not included in the header of Transfer Frames.

The TC Space Data Link Protocol uses an optional identifier, called the Multiplexer Access Point Identifier (MAP ID), that is used to create multiple streams of data within a Virtual Channel. All the Transfer Frames on a Virtual Channel with the same MAP ID constitute a MAP Channel. If the MAP ID is used, a Virtual Channel consists of one or multiple MAP Channels.

Figure 3-1 shows the relationship among the channels of the TM, TC, and AOS Space Data Link Protocols.

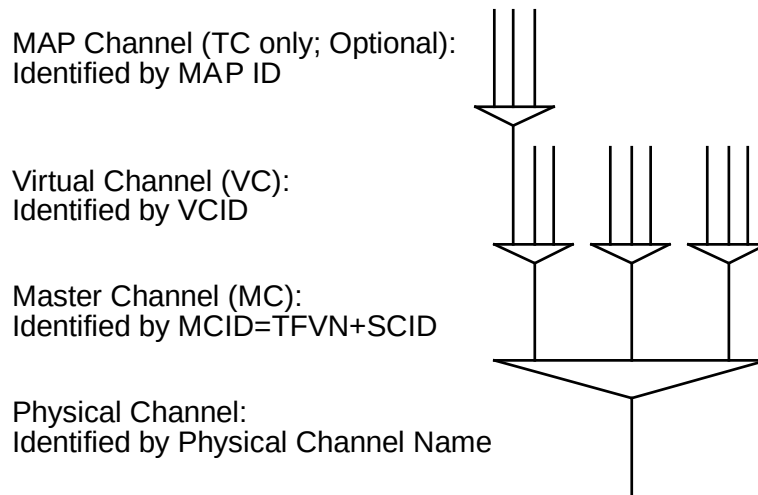


Figure 3-1: Relationships between Channels of the Space Data Link Protocols

The Proximity-1 Space Link Protocol—Data Link Layer uses a triad of multiplexing capabilities, which is incorporated for specific functionality within the link. The Spacecraft Identifier (SCID) identifies the source or destination of Transfer Frames transported in the link connection based upon the Source-or-Destination Identifier. The Physical Channel Identifier (PCID) provides up to two independently multiplexed channels. The Port ID provides the means to route user data internally (at the transceiver's output interface) to specific logical ports, such as applications or transport processes, or to physical ports, such as onboard buses or physical connections (including hardware command decoders).

Table 3-1 summarizes the identifiers of the Space Data Link Protocols. The values of these IDs are maintained by the SANA registries (reference [47]).

Table 3-1: Identifiers of Space Data Link Protocols

Identifiers	TM Space Data Link Protocol	TC Space Data Link Protocol	AOS Space Data Link Protocol	Proximity-1 Space Link Protocol—Data Link Layer
Transfer Frame Version Number (TFVN)	Always 1 (binary encoded number is 00)	Always 1 (binary encoded number is 00)	Always 2 (binary encoded number is 01)	Always 3 (binary encoded number is 10)
Spacecraft Identifier (SCID)	0 to 1023	0 to 1023	0 to 255	0 to 1023
Physical Channel Identifier (PCID)	N/A	N/A	N/A	0 to 1
Virtual Channel Identifier (VCID)	0 to 7	0 to 63	0 to 63	N/A
Multiplexer Access Point Identifier (MAP ID)	N/A	0 to 63	N/A	N/A
Port Identifier	N/A	N/A	N/A	0 to 7

3.2.3 SERVICES PROVIDED BY DATA LINK PROTOCOLS

The Space Data Link Protocols provide several services to transfer a variety of data on a space link. The most important service is a service to transfer variable-length data units known as packets (i.e., protocol data units of protocols of the Network Layer). In addition to this service, the Space Data Link Protocols provide services to transfer fixed- or variable-length data units with private (non-CCSDS) formats, short fixed-length data units for reporting on real-time functions, and bit streams.

Table 3-2 shows a summary of the services provided by the TM/TC/AOS Space Data Link Protocols categorized by the types of data transferred by the services. For complete definition of these services, refer to references [5], [6], and [7].

NOTE – The Proximity-1 Space Link Protocol—Data Link Layer is not included in this table because no formal service definition is given in the Recommended Standard (references [18]). In spite of not having a formal service definition, Proximity-1 can deliver the same SDUs delivered by the Packet Service, Encapsulation Service, and the TC VCA Service.

Table 3-2: Summary of Services Provided by Space Data Link Protocols

Type of Service Data Units	TM Space Data Link Protocol	TC Space Data Link Protocol	AOS Space Data Link Protocol
Packets (NOTE 1)	Packet Service, Encapsulation Service	MAP Packet Service, VC Packet Service, Encapsulation Service	Packet Service, Encapsulation Service
Fixed-length private data	VC Access Service	(None)	VC Access Service
Variable-length private data	(None)	MAP Access Service, VC Access Service	(None)
Short fixed-length data	VC FSH Service (NOTE 2), MC FSH Service, VC OCF Service (NOTE 3), MC OCF Service	(None)	Insert Service, VC OCF Service (NOTE 3)
Bit stream	(None)	(None)	Bitstream Service
Transfer Frames	VC Frame Service, MC Frame Service	VC Frame Service, MC Frame Service	VC Frame Service, MC Frame Service

NOTES

- 1 Packets directly transferred by the Space Data Link Protocols must have Packet Version Numbers authorized by CCSDS. These Packet Version Numbers are found in reference [28]. Other packets can be transferred using the Encapsulation Service defined in reference [29].
- 2 FSH = Frame Secondary Header.
- 3 OCF = Operational Control Field.

3.2.4 SYNCHRONIZATION AND CHANNEL CODING

The standards of the Synchronization and Channel Coding Sublayer provide some additional functions necessary for transferring Transfer Frames over space links. These functions are delimiting/synchronizing Transfer Frames, error-correction coding/decoding, and bit transition generation/removal. CCSDS has five standards for Synchronization and Channel Coding:

- a) A set of three TM specifications:
 - 1) *TM Synchronization and Channel Coding* (reference [8]),
 - 2) *Flexible Advanced Coding and Modulation Scheme for High Rate Telemetry Applications* (reference [49]), or ‘SCCC’,

- 3) *CCSDS Space Link Protocols over ETSI DVB-S2 Standard* (reference [50]), or ‘DVB-S2’;
- b) *TC Synchronization and Channel Coding* (reference [9]);
- c) *Proximity-1 Space Link Protocol—Coding and Synchronization Sublayer* (reference [19]).

The three TM specifications define alternative synchronization and channel coding schemes used with the TM or AOS Space Data Link Protocol, *TC Synchronization and Channel Coding* defines synchronization and channel coding schemes used with the TC Space Data Link Protocol, and *Proximity-1 Space Link Protocol—Coding and Synchronization Sublayer* defines synchronization and channel coding schemes used with the Proximity-1 Space Link Protocol.

Options a)2) and a)3) are recommended only for high rate downlink. *TM Channel Coding Profiles* (reference [51]) provides suggestions on how to accomplish channel coding options.

Table 3-3 summarizes the functions and schemes provided by the Synchronization and Channel Coding standards.

Table 3-3: Functions of Synchronization and Channel Coding Standards

Functions	TM Synchronization and Channel Coding	TC Synchronizatio n and Channel Coding	Proximity-1 Space Link Protocol— Coding and Synchronization Layer
Error Correction Scheme + Frame Validation (see NOTE 3)	- Convolutional Coding + FECF; - Reed Solomon Coding; - Concatenated Coding; - Turbo Coding + FECF ; - Low-density Parity-check (LDPC) Coding; - SCCC + FECF; - DVB-S2 + FECF	- BCH Coding - BCH Coding + FECF	- Convolutional Coding + Attached CRC; - Low-density Parity-check (LDPC) Coding + Attached CRC;
Pseudo- Randomizati on (see NOTE 3)	Cyclic Pseudo- noise Sequence*	Cyclic Pseudo- noise Sequence*	Cyclic Pseudo- noise Sequence (Used – mandatorily- only for LDPC Coding)
Frame Synchronizat ion	32-bit (or longer) Attached Sync Marker (ASM) depending on applied coding scheme	16-bit Start Sequence	24-bit Attached Sync Marker

NOTES

- 1 ‘*’ in the table denotes an optional function.
- 2 When a box of the table shows several options, only one option can be applied at a given time.
- 3 When only an Error Correction scheme is mentioned, it means that the scheme is also capable of validating the frame; i.e. declaring it erroneous or error free. In other cases, for TM/TC/AOS a Frame Error Control Field is used for error detection while Proximity-1 uses a Cyclic Redundancy Code (CRC) attached to the frame (but not part of the frame). The Frame Error Control Field is defined in the Recommended Standards on the TM/TC/AOS Space Data Link Protocols, and not in the Recommended Standards on Synchronization and Channel Coding.
- 4 The Cyclic Pseudo-noise Sequence used by TM Synchronization and Channel Coding differs from that one used for both TC Synchronization and Channel Coding and Proximity-1 Space Link Protocol—Coding and Synchronization Layer.

Summaries of concept and rationale for TM Synchronization and Channel Coding, TC Synchronization and Channel Coding, and Proximity-1 Space Link Protocol—Coding and Synchronization Layer are contained in references [33], [34], and [32], respectively.

3.3 NETWORK LAYER

3.3.1 GENERAL FEATURES OF NETWORK PROTOCOLS

CCSDS has one service for interfacing at the Network Layer: the Encapsulation Service. Within this service, there are two different types of packets: Space Packets defined in the Space Packet Protocol (reference [4]) and Encapsulation Packets defined in the Encapsulation Service (reference [29]). It should be noted that IP over CCSDS (reference [45]) exclusively utilizes the Encapsulation Packet.

NOTE – While the Space Packet protocol provides a service interface that could be used for access to the network layer, the service is subsumed by the Encapsulation service.

The Space Packet Protocol was developed to transfer data (1) from a source on a spacecraft to one or multiple destinations on the ground or on (an)other spacecraft, or (2) from a source on the ground to one or multiple destinations on one or multiple spacecraft. When protocol data units of this protocol traverse the data system of a space mission (i.e., onboard networks, onboard data handling system, ground stations, control centers), the application identifier (APID) that is part of each packet is used for determining the path that packet will take. All decisions about how packets are to be handled and forwarded, based on this APID, are set by management agreement and are not a formal part of the protocol. There should be no expectation of interoperable handling of APIDs and paths in a cross support situation unless agreements have been clearly defined as to how such forwarding is to be done.

The Space Packet Protocol provides the capability to transfer space application data over a path that involves a ground-to-space or a space-to-space communications link.

By using the Encapsulation Service, other CCSDS-recognized Network Protocols such as Delay Tolerant Networking (DTN) (references [55] and [56]) and IP can be used over space links.

The protocol data units of the Space Packet Protocol are called Space Packets, while the protocol data units of IP are called IP datagrams.

SPP and IP do not provide any QoS mechanisms for reliable delivery, in-order delivery, or duplicate suppression. If these functions are required they must be implemented by a higher-layer (e.g. transport layer) protocol.

Over a space link, protocol data units of the network protocols (i.e. DTN, IP, or others carried through SPP or Encapsulation Service) are transferred within the Space Data Link Protocols.

The Space Data Link Protocols have the capability to carry several protocol data units of the Internet Protocol, multiplexed or not-multiplexed, within the Encapsulation packet. IP over CCSDS (reference [45]) specifies how CCSDS-recognized IP datagrams are transferred over the link.

3.3.2 ADDRESSING OF NETWORK PROTOCOLS

Two types of addresses are used by the Network Layer protocols: Path Address and End System Address.

The Path Address, used by the Space Packet Protocol, identifies a Logical Data Path (LDP) in the network from a source to one or multiple destinations. Path Addresses are not addresses as defined in reference [2], but CCSDS has been using Path Addresses as a way of identifying logical connections in the Space Packet Protocol (reference [4]). Configuration of LDPs is done by management activities relying on specific user to provider agreements and mission/subnetwork-specific mechanisms, and not by any explicit features of this protocol.

An End System Address, as used by IP and DTN, unambiguously identifies a single end system or a group of end systems. If it is necessary to identify both the source and destination when using End System Addresses, a pair of End System Addresses must be used. These addresses are specified in the IP or DTN PDUs and they are used by the IP or DTN routing nodes to perform routing decisions at each step along the end-to-end path.

As already mentioned, CCSDS Encapsulation Service allows the use of other CCSDS recognized Network Protocols within their own addressing types.

CCSDS is developing Delay Tolerant Networking (references [55] and [56]) as the means to perform interoperable internetworking in space, in either disrupted or delayed end-to-end communication environments.

3.4 TRANSPORT LAYER

CCSDS has developed the SCPS Transport Protocol (SCPS-TP) (reference [13]) for the Transport Layer. The CCSDS File Delivery Protocol (CFDP) (reference [15]) also provides the functionality of the Transport Layer, but it provides some functions (i.e., functions for file management) of the Application Layer as well.

SCPS-TP supports end-to-end communications between applications and is designed to meet the needs of a broad range of space missions. It defines extensions to TCP and incorporates UDP by reference. It may be used on top of the Space Packet, Encapsulation Packet, or IP over CCSDS.

CFDP provides the functionality of the Application Layer (i.e., functions for file management), but it also provides functions of the Transport Layer.

Transport protocols used in the Internet (such as TCP, reference [24], and UDP, reference [25]) can also be used on top of the Encapsulation packet, or IP over CCSDS space links.

IPSec (reference [27]) can be used with the Internet Protocol suite to provide end-to-end data protection capability.

3.5 APPLICATION LAYER

CCSDS has developed five protocols for the Application Layer:

- a) Asynchronous Messaging Service (AMS) (reference [46]);
- b) CCSDS File Delivery Protocol (CFDP) (reference [15]);
- c) Lossless Data Compression (reference [16]);
- d) Image Data Compression (reference [17]);
- e) Lossless Multispectral & Hyperspectral Image Compression (reference [48]).

The Asynchronous Messaging Service (AMS) implements an interoperable protocol under which the mission modules, distinct sequential flows of application control logic, whether called processes, tasks, or threads, each one producing and consuming mission information, may be designed without explicit awareness of which other modules are currently operating nor of where they are deployed.

CFDP is designed to meet the needs of space missions to transfer files. It is a file transfer protocol, but it also provides services typically found in the Transport Layer, i.e. complete, in-order, without duplicate data delivery. It can be used on top of any protocol of the Network Layer (e.g., Space Packet Protocol, Encapsulation Service, IP over CCSDS), or directly on top of the TC Space Data Link Protocol or Proximity-1 Space Link Protocol if a Virtual Channel, a MAP, an APID, or a Port is dedicated to CFDP. In some circumstances it can be used on top of UDP, TCP or SCPS-TP. A summary of concept and rationale of CFDP is contained in reference [35].

The Lossless Data Compression standard was developed to increase the science return as well as to reduce the requirement for onboard memory, station contact time, and data archival volume. This standard guarantees full reconstruction of the original data without incurring any distortion in the process. It is intended to be used together with the Space Packet Protocol or CFDP. A summary of concept and rationale of Lossless Data Compression is contained in reference [36].

The Image Data Compression standard was developed to establish a standard for a data compression algorithm applied to digital image two-dimensional spatial data from payload instruments. With this standard, quantization or other approximations used in the compression process may result in the inability to reproduce the original data set without some distortion. It is intended to be used together with the Space Packet Protocol, CFDP, or the AOS Space Data Link Protocol.

Applications protocols used in the Internet can be used over TCP (with or without the SCPS-TP extensions) or UDP as long as the underlying links are sufficiently short (<1 sec) and continuously available. Typically, an application is written to use the reliable stream-oriented service of TCP *or* the unreliable datagram service of UDP, but not both. Some exceptions to this exist, however, in which applications are written to operate over either service.

The Lossless Multispectral & Hyperspectral Image Compression standard provides a data compression algorithm applied to digital three-dimensional image data from payload instruments, such as multispectral and hyperspectral imagers, and to specify the compressed data format.

Each project (or Agency) may elect to use application-specific protocols not recommended by CCSDS to fulfill their mission requirements in the Application Layer over CCSDS space communications protocols.

4 EXAMPLES OF PROTOCOL CONFIGURATIONS

4.1 GENERAL

This section shows some examples of how space communications protocols of various layers are used in space data systems.

Five examples of protocol configurations are shown in this section. There are many other combinations of protocols that can be used in space data systems, but it is not the intention of this Report to enumerate all possible combinations of protocols. The following examples are selected to illustrate the basic functionality of the space communications protocols.

For each example in this section, two diagrams are shown. The first diagram shows a stack of protocols used over a space link (i.e., a link between a spacecraft and a ground station or between two spacecraft).

A space data system consists of one or more onboard subnetworks, one or more space links, and one or more ground subnetworks. In this section, however, a simple space data system consisting of four major elements (see figure 4-1) is used to illustrate how space communications protocols are used in an end-to-end space data system. It will be shown that some space communications protocols are used for end-to-end communications between onboard and ground end systems, and some space communications protocols are used only for communications over the space link.

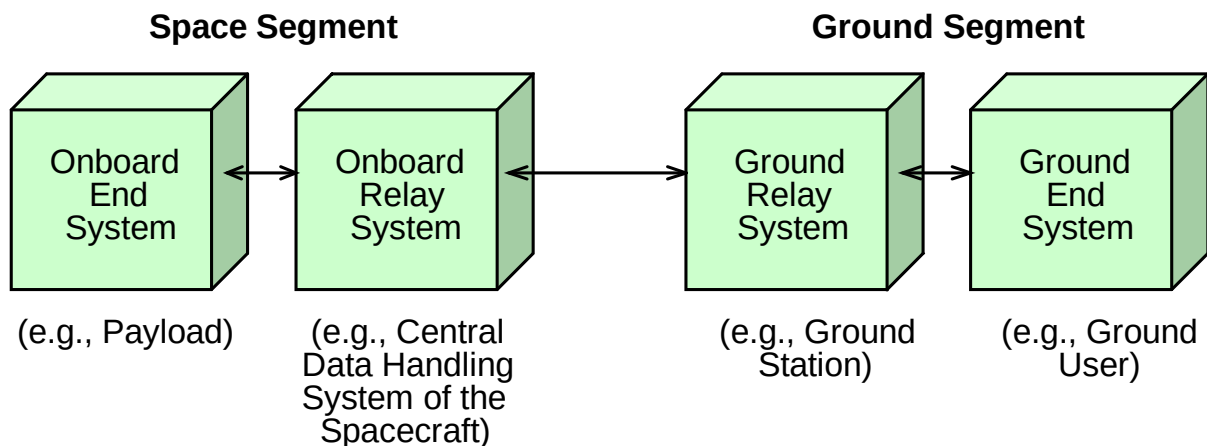


Figure 4-1: Simple Space Data System Model

The primary difference among the five examples shown in this section is the selection of the protocol used for end-to-end routing or forwarding. In a space data system, user data traverse subnetworks (i.e., one or more onboard subnetworks, one or more space links, and one or more ground subnetworks). One of the protocols used in a space data system provides the capability of routing user data from a source to a destination through these subnetworks.

This functionality is called ‘end-to-end routing’ in this Report (see definitions of ‘routing’ and ‘forwarding’ in 1.3.2).

The following protocols are used for end-to-end routing in the following examples:

- a) Space Packet Protocol or Encapsulation Service;
- b) IP over CCSDS over the Encapsulation Packet;
- c) CFDP.

NOTES

- 1 In the following figures, ‘Prox Space Data Link Protocol’ denotes the Proximity-1 Space Link Protocol—Data Link Layer.
- 2 In the following figures, the Synchronization and Channel Coding standards are omitted for simplicity reasons.
- 3 CCSDS is developing DTN (references [55] and [56]) as the means to internetwork in space.

4.2 END-TO-END FORWARDING USING PACKETS DEFINED BY CCSDS

In this example, the Space Packet is used for end-to-end forwarding. The Space Packet Protocol was designed by CCSDS to meet the requirements of space missions for efficient transfer of processed data over space links. This configuration is suited to space missions that require the simple APID tag and forwarding capabilities provided by the Space Packet Protocol.

Figure 4-2 shows an example of protocol configuration on a space link, and figure 4-3 shows an example of protocol configuration in an end-to-end space data system. At each intermediate system some mechanism, not specified in the SPP protocol, examines the APID and forwards the data to the next node that it has been instructed to use. There is no endpoint address and there is no specified mechanism for doing this interoperably. It is done by management and external agreement between user and service provider.

When the Space Packet Protocol is used for end-to-end forwarding, in the ground subnetwork Space Packets are usually transferred with a Space Link Extension (SLE) Service (see references [38]–[42]).

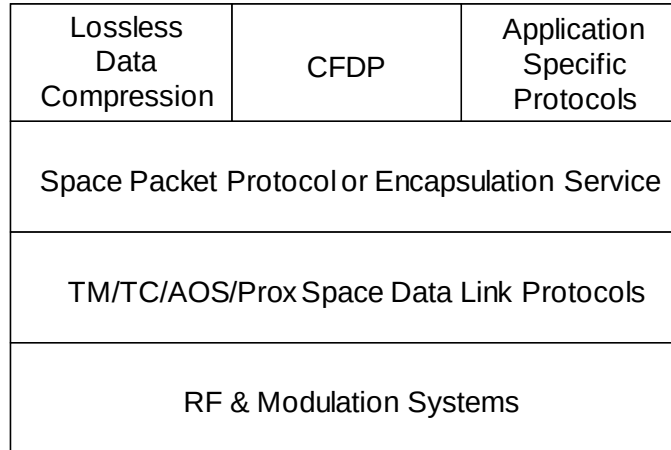


Figure 4-2: Protocol Configuration on a Space Link When Space Packet Protocol or Encapsulation Service Is Used for End-to-End Forwarding

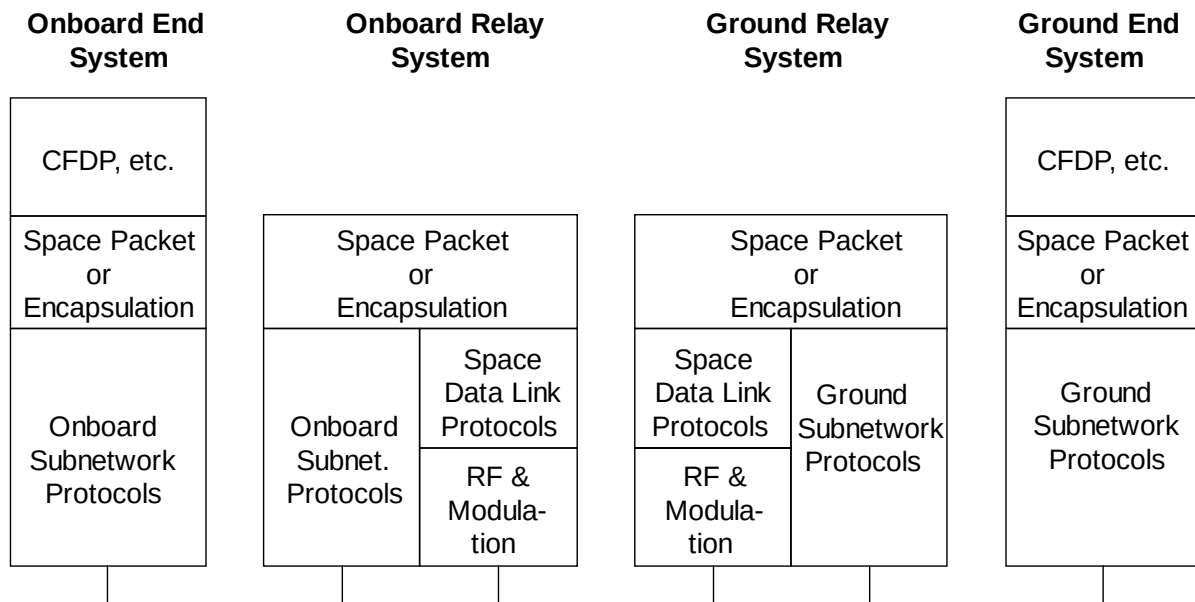


Figure 4-3: Protocol Configuration in a Space Data System When Space Packet Protocol or Encapsulation Service Is Used for End-to-End Forwarding

4.3 IP OVER CCSDS FOR END-TO-END ROUTING

In the fourth example, one of the CCSDS recognized IP datagrams defined in SANA is used for end-to-end routing. This configuration is suited to space missions that require integration of their space segments into the Internet when end-to-end internetworking is required and when connectivity and RTL T is suitable to support this approach.

Figure 4-4 shows an example of protocol configuration on a space link, and figure 4-5 shows an example of protocol configuration in an end-to-end space data system.

Protocol data units (datagrams) of IP are transferred by Space Data Link Protocols using the IP over CCSDS protocol in order for the Space Data Link Protocols to process IP datagrams efficiently.

In this example, it is assumed that the Internet is directly extended into the space segment. Most Internet end-to-end protocols and SCPS-TP can be used on top of IP. SCPS-TP can be converted to TCP/UDP at a relay system.

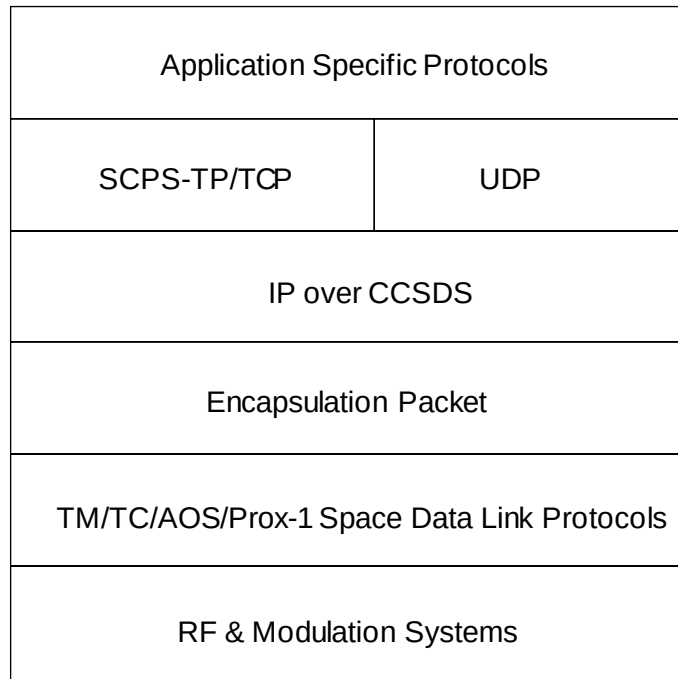


Figure 4-4: Protocol Configuration on a Space Link When IP over CCSDS Is Used for End-to-End Routing

At each intermediate system in an IP deployment a routing mechanism, specified in the IP protocol, examines the destination address and makes a routing decision that sends the data to the next node in the route. The endpoint address is explicit and all of the mechanisms for doing this interoperably are fully specified. The encapsulation packet provides the shim to insert the IP datagrams into a CCSDS space link and to extract it at the other end.

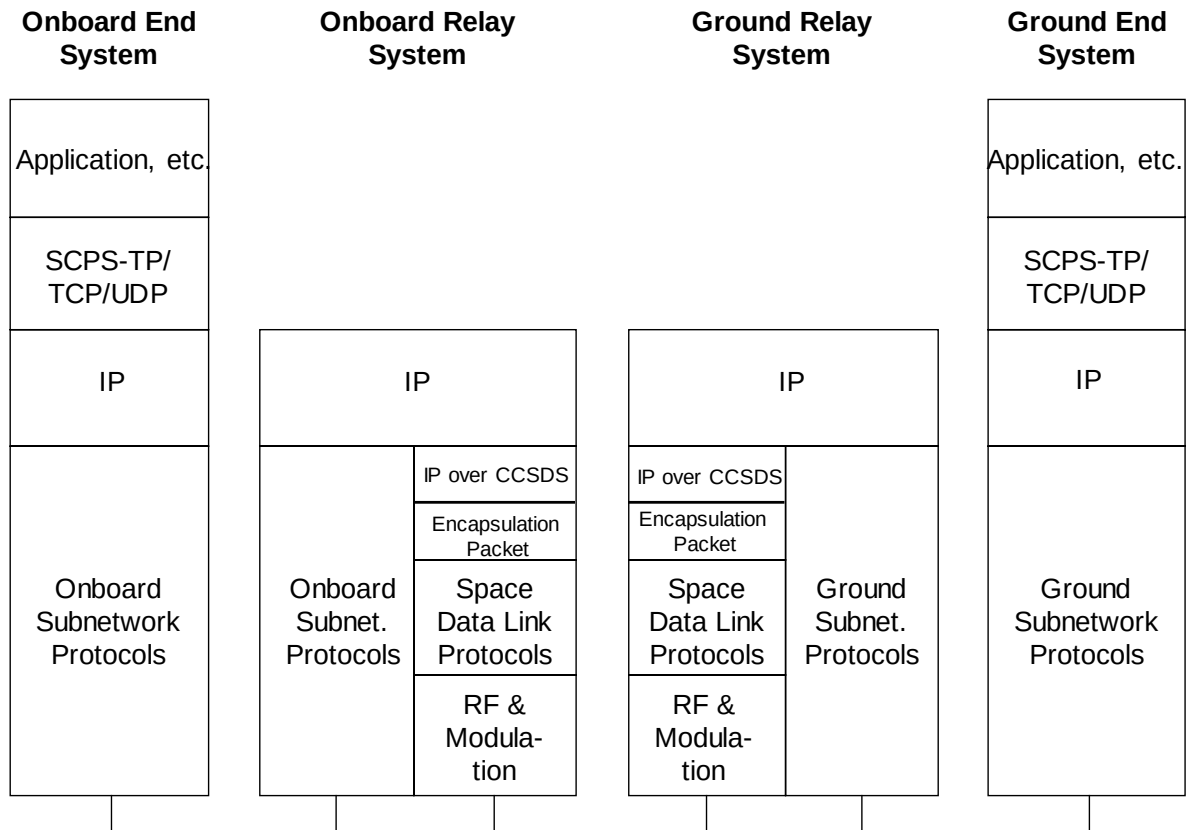


Figure 4-5: Protocol Configuration in a Space Data System When IPoC Is Used for End-to-End Routing

4.4 CFDP FOR END-TO-END FORWARDING

In the final example, CFDP is used for end-to-end forwarding. CFDP is a file transfer protocol, but it also has the capability to forward files through a space data system. This configuration is suited to space missions in which most data are transferred as files. In most deployments of CFDP the forwarding features defined in the Class 3 and 4, or Store and Forward Overlay (SFO) options are not used, but they may be. Typical deployments use only the CFDP Class 1 (unreliable) or Class 2 (reliable) modes and deploy CFDP protocol engines only in the end nodes.

Figure 4-6 shows an example of protocol configuration on a space link, and figure 4-7 shows an example of protocol configuration in an end-to-end space data system.

In this example, it is assumed that protocol data units of CFDP are carried either by the Space Packet Protocol or the Encapsulation Service over the space link, but they can also be carried by IP over CCSDS.

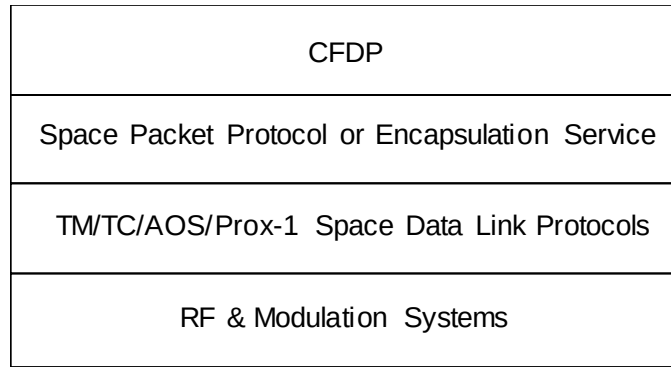


Figure 4-6: Protocol Configuration on a Space Link When CFDP Is Used for End-to-End Forwarding

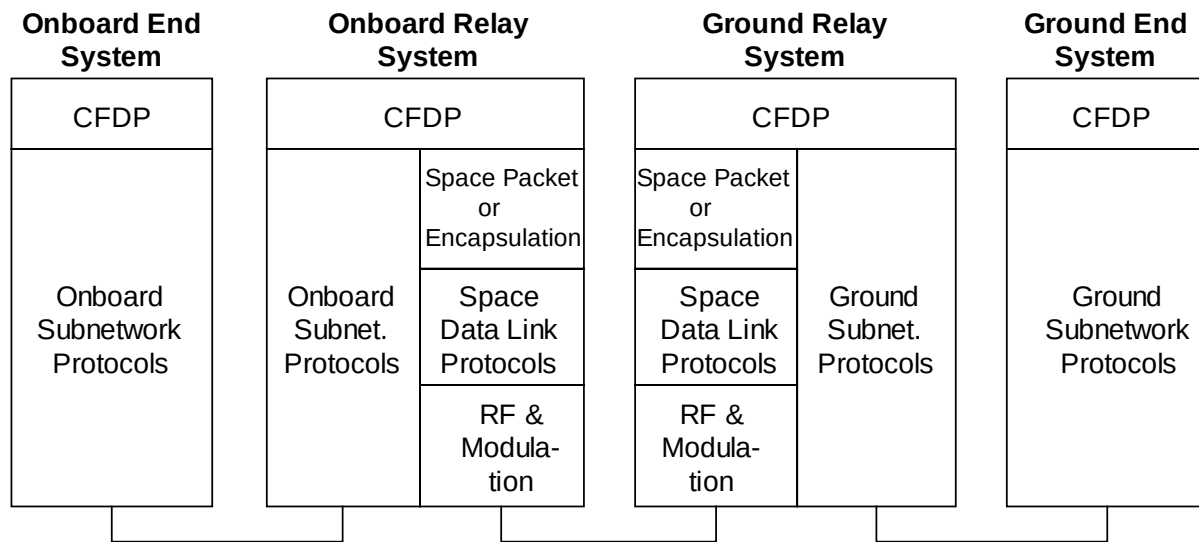


Figure 4-7: Protocol Configuration in a Space Data System When CFDP Is Used for End-to-End Forwarding

Figure 4-7, which represents the use of multiple CFDP protocol engines and the Store and Forward Overlay (SFO), is an example of end-to-end forwarding. At each intermediate system the CFDP protocol is terminated, the file is re-assembled, and the SFO procedure is used to forward the data to the next node that it has been instructed to use. This is hop by hop file delivery. It is done by management and external agreement between user and service provider.

CCSDS is developing the DTN networking protocols (see references [55] and [56]) that are suitable for use in both near Earth, connected, environments and in deep space, or disconnected, environments with long RTLT. The recommended approach for providing end-to-end file services with intermediate hops is to utilize CFDP over DTN. In this configuration the DTN networking protocol manages the end-to-end routing and delivery of the CFDP PDUs.

ANNEX A

ACRONYMS

This annex lists the acronyms and abbreviations used in this Report.

AOS	Advanced Orbiting Systems
APID	Application Process Identifier
CCSDS	Consultative Committee for Space Data Systems
CFDP	CCSDS File Delivery Protocol
DTN	Delay Tolerant Networking
FSH	Frame Secondary Header
FTP	File Transfer Protocol
ID	Identifier
IP	Internet Protocol
LDP	Logical Data Path
MAP	Multiplexer Access Point
MC	Master Channel
MCID	Master Channel Identifier
N/A	Not Applicable
OCF	Operational Control Field
PCID	Physical Channel Identifier
PVN	Packet Version Number
QoS	Quality of Service
SANA	Space Assigned Numbers Authority
SCID	Spacecraft Identifier
SCPS	Space Communications Protocol Standards
SCPS-FP	Space Communications Protocol Standards File Protocol
SCPS-NP	Space Communications Protocol Standards Network Protocol

SCPS-SP	Space Communications Protocol Standards Security Protocol
SCPS-TP	Space Communications Protocol Standards Transport Protocol
SLE	Space Link Extension
TC	Telecommand
TCP	Transmission Control Protocol
TDM	Time Division Multiplexing
TFVN	Transfer Frame Version Number
TM	Telemetry
UDP	User Datagram Protocol
VC	Virtual Channel
VCID	Virtual Channel Identifier